

Communication for Development
23MCOM19C
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Draft Submission

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Why did we pick this topic? - What does the Campaign aim to achieve?

Behavioural Change

Secondary Research

Nowadays, many people use cooking oil widely, for example in restaurants or at home, but throwing away cooking oil can cause many problems, including the possibility of clogged drains, leading to pollution and a negative impact on the environment. Some countries have passed laws for the safe disposal of oils in order to protect the environment (Daud et al., 2020). Considering that (WCOs) are an essential member of the family of used vegetable oils (UVOs), they received great attention due to their abundance, which exceeded 190 million metric tons, and it was necessary to benefit from them in various industries such as lubricants, animal feed, asphalt additives, biofuels, This would be a good solution to solve the fuel problem in rural areas (Mannu et al., 2020).

According to Loizides et al. (2019), Recent research has indicated a large global production of UCO which refers to used cooking oil, which can be reused in the manufacture of soap, biofuel and other industries. The production reaches 16.54 million tons annually from various regions in the world, including China, Malaysia, the United States, Europe and other countries. Europe alone produces 1.66 million tons of UCO annually. The ban imposed on the use of UCO as animal feed has led to trends in the European Union since 2002 encouraging renewable energy, and this has led to the growth of the UCO market for the production of biofuels. However, domestic UCO production may still be largely underutilized due to logistical challenges, leading to inequitable and inappropriate disposal of UCO and also resulting in environmental damage and economic burdens. To address this matter, new methods have been developed for example InnovOleum reverse logistics system with the aim of collecting the material and recycling it properly to provide social, environmental and economic benefits, including green initiatives in schools.

According to Naima and Liazid (2013), Millions of tons of waste oil are disposed of around the world by burying it in the ground or throwing it into the drain, In some countries, such as the United States, it is illegal to dispose of used cooking oil. It must be disposed of using paper towels or absorbent materials such as cat litter, or by freezing methods, because it may cause blockage of the drain through the accumulation of solids and their growth to a huge size, which ultimately leads to... Clogged waste and sewage pipes, and not only that, sewage can reach bathtubs, toilets, and floor drains (Lehman College)

Naima and Liazid (2013) pointed to Waste oil is considered an important commodity and has a major impact because it is used in many industries, including the manufacture of fertilizers, animal feed, some types of soap, and also cosmetics for women and other products, the most important of which is the production of diesel fuel. This was discovered through the increased demand for Energy and the desire to find alternatives to engines, and also noting the seriousness of the environmental damage that the world faces due to the disposal of waste oils, scientists analysed the waste oils and compared them to petroleum products until they

discovered characteristics similar to those of biodiesel fuel, which is a combustible and biodegradable fuel that is safe and does not cause danger. It is made from animal fats as well as vegetable oils. The raw materials for biodiesel come from renewable local sources, and diesel is considered cleaner-burning than petroleum-based fuel.

References

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Loizides, M. I., Loizidou, X. I., Orthodoxou, D. L., & Petsa, D. (2019). Circular bioeconomy in action: collection and recycling of domestic used cooking oil through a social, reverse logistics system. *Recycling*, 4(2), 16. <https://www.mdpi.com/2313-4321/4/2/16>

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Lehman College. Used Cooking Oil Recycling. <https://www.lehman.edu/administration/sustainability/cooking-oil.php>

Primary Research Survey - 68 respondents

Oil to Oil Recycling (Responses)

Gender Analysis:

- The data consisted of responses from 51% females and 49% males.
- There did not appear to be a significant difference in the responses based on gender for most questions.

Occupation Analysis:

- 76% of respondents were university students from various faculties such as computer science, business administration, mass communication, psychology, and others.

- 24% were professionals including 3% bankers, 3% journalists, 3% instructional designers, 3% software engineers, 3% customer service representatives, and others.
- The responses did not vary significantly based on occupation, except for a few instances where students may have different perspectives or priorities compared to professionals.

Household Income Analysis:

- 55% of respondents preferred not to disclose their household income.
- Among those who provided income data, 18% had an income under 25,000 EGP, 15% between 25,000-49,999 EGP, 12% between 50,000-74,999 EGP, 9% between 75,000-99,999 EGP, and 9% had 100,000 EGP or more.
- There did not appear to be a clear pattern in the responses based on household income, suggesting attitudes towards cooking oil recycling may not be strongly influenced by income levels.

Age Analysis:

- 85% of respondents with age data were between 17-25 years old, as many were university students.
- There was insufficient data to analyze responses based on a broader age range.

Social Media Usage:

- 73% reported checking social media very frequently (4-5 on the scale), indicating high engagement.
- There was a potential correlation between high social media usage and awareness or interest in cooking oil recycling, as social media could spread information.

Cooking Oil Usage and Recycling:

- 55% used 1-2 liters of cooking oil per month, 21% less than 1 liter, 15% used 3-4 liters, and 9% more than 4 liters.
- 70% had not previously attempted to clean and reuse cooking oil themselves, while 30% had (with 21% doing so rarely).
- 36% preferred cleaning and reusing cooking oil themselves, 33% having it collected from home, 18% dropping it off at a recycling center, and 13% had no preference.
- 52% believed recycling cooking oil is environmentally beneficial, 15% did not, and 33% were unsure.
- 45% were willing to pay a small fee for a cooking oil recycling service, 39% were maybe willing, and 15% were not willing.
- 58% thought the government should provide incentives for recycling cooking oil.
- 61% would trust oil that has been cleaned and reused.
- 64% thought there should be more public awareness about recycling cooking oil.
- 55% would encourage friends and family to recycle cooking oil.

The data provides valuable insights from a diverse group of respondents. While there were some variations based on demographics, the general trend suggested growing interest and awareness regarding the environmental benefits of cooking oil recycling, as well as a

willingness to participate with proper infrastructure and incentives. High social media usage presents an opportunity for raising awareness on this issue.